

YUCHENG YIN (阴煜程)

✉ yyin@caltech.edu

🔗 cuberyyc808.github.io

(last updated 2026-09-26)

EDUCATION

California Institute of Technology

Ph.D. in Physics

Pasadena, CA, USA

09/2026 – Present

School of Physics, Peking University

Bachelor's Degree of Science

Beijing, China

09/2021 – 07/2025

- Major in Physics (Astrophysics)
- Research interests: gravitational wave astrophysics, black hole perturbation theory.
- Thesis: Gravitational radiation of binary extreme-mass-ratio inspiral systems (Click [here](#))

RESEARCH EXPERIENCE

Kavli Institute for Astronomy and Astrophysics, Peking University

Undergraduate researcher

Beijing, China

02/2023 – 06/2025

Project: Study the three-body dynamics and GW radiation of binary extreme-mass-ratio inspiral (b-EMRI)

Supervisor: Prof. [Xian Chen](#)

Department of Physics, National University of Singapore

Summer intern

Singapore

06/2024 – 08/2024

Project: Waveform modeling of b-EMRI systems using black hole perturbation theory

Supervisor: Prof. [Alvin J. K. Chua](#)

Niels Bohr Institute, University of Copenhagen

Visiting student

Copenhagen, Denmark

10/2025 – 03/2026

Project: Inhomogeneous Sasaki-Nakamura formalism

Supervisor: Dr. [Rico K. L. Lo](#)

SKILLS

- Programming Languages: Mathematica ~ Julia ~ Python > C
- BHPT: RWZ formalism, Teukolsky formalism, Sasaki-Nakamura formalism, QNMs theory, Self-force, ...
- Others: Numerical methods to solve ODE and PDE

PROFESSIONAL MEMBERSHIP

[LISA](#) (Laser Interferometer Space Antenna) consortium, **Core** member

2025 – Present

Waveform working group - working on EMRI OPA flux dataset generation and validation

TOOLS AND CODES

- `KerrGeodesics.jl`: Solves timelike Kerr geodesics.
- `AdaptiveLevin.jl`: Evaluates one- and two-dimensional highly oscillatory integrals.
- `GeneralizedSasakiNakamura.jl`: Solves the radial Teukolsky equation.
- `SpinWeightedSpheroidalHarmonics.jl`: Solves the angular Teukolsky equation.

TALKS

Contributed talks

BIMSA Workshop on Gravitational Wave Astronomy, Beijing 04/2025

Invited talks

School of Physics and Astronomy, Sun Yat-sen University 11/2025

HONORS

Linbridge Scholarship, funded by Prof. Douglas N. C. Lin 2024

Outstanding Undergraduate Thesis from Peking University 2025

REFERENCES

Vitor Cardoso

Professor, University of Copenhagen & Universidade de Lisboa

Alvin J. K. Chua

Assistant Professor, National University of Singapore

Xian Chen

Associate Professor, Peking University